

SOC/L1 -11.1. Malware Classification

Task 1 - Introduction

Malware refers to software or code created to damage systems, steal information, or allow unauthorised access. It appears in many forms, each with different tactics and goals. Malware can affect businesses in many ways, from financial loss and stolen data to disrupted services and reputational damage.

Task 2 - Malware Types

Category	Main Purpose	Typical Behaviour	Real-World Example
Adware	Display unwanted advertisements	Display unwanted advertisements	Fireball(opens in new tab) .
Spyware	Collect information secretly	Tracks browsing, records keystrokes, captures data	Hermit(opens in new tab) .
Ransomware	Extort money by locking files or systems	Encrypts data, displays ransom note	WannaCry(opens in new tab) .
Wiper	Destroy data or systems	Overwrites or deletes files permanently	PathWiper(opens in new tab) .
C2 (Command & Control)	Enable remote control by an attacker	Connects to remote server, receives commands	Emotet(opens in new tab) .
Data stealer	Steal sensitive information	Exfiltrates files, credentials, or documents	Lumma Stealer(opens in new tab) .
Keylogger	Record user keystrokes	Captures everything typed on keyboard	Zeus(opens in new tab) .
Cryptominer	Use resources to mine cryptocurrency	High CPU usage, slows systems, network traffic	Coinhive

Below are some simple scenarios that show how this type of malware might appear within an organisation.

- **Adware:** A user's browser starts showing pop-up ads every few minutes, even when no websites are open. Security tools identify a suspicious program running in the background, which turns out to be adware.
- **Spyware:** An employee notices their personal emails have been accessed by someone else. Forensics show a spyware program on the workstation, which was recording keystrokes and sending the information to an attacker.
- **Ransomware:** Several computers display a ransom note, saying that all files have been encrypted and payment is required to unlock them. The traces the incident to a ransomware attack triggered by a malicious email attachment.
- **Wiper:** An organisation's servers suddenly crash, and files are overwritten with random data. Investigation reveals wiper malware designed to destroy data rather than demand a ransom.
- **Command and Control (C2) malware:** An endpoint is found reaching out to unfamiliar domains at odd hours. Analysts discover malware that gives attackers remote access, allowing them to run commands or move files.
- **Data stealer:** Sensitive documents start leaking online. The incident response team finds a data stealer malware that collected files from user directories and sent them to an external server.
- **Keylogger:** Users report unauthorised transfers from company bank accounts. Analysis shows a running on the accountant's computer, recording every keystroke, including passwords.
- **Cryptominer:** Several desktops become slow and fans are running constantly. Monitoring tools reveal high usage linked to a cryptominer that hijacks the computers to mine cryptocurrency.

Task 3 - Real-World Malware Examples

Spyware: Case Study, [Pegasus\(opens in new tab\)](#): Pegasus is one of the most well-known spyware tools, used to target mobile phones worldwide. Attackers send a specially crafted message or exploit a vulnerability to install Pegasus silently.

Ransomware: Case Study, [Akira\(opens in new tab\)](#): Akira is a modern ransomware variant known for targeting businesses, schools, and public services. Attackers use phishing emails, stolen credentials, or remote access software to breach networks.

Wiper: Case Study, [Shamoon\(opens in new tab\)](#): Shamoon targeted energy companies in the Middle East, most famously Saudi Aramco. The malware rapidly overwrote files with useless data, making systems unusable.

Data Stealer: Case Study, [Agent Tesla\(opens in new tab\)](#): Agent Tesla is commonly delivered by phishing emails with attachments. Once opened, it captures keystrokes, screenshots, and credentials from web browsers and email clients.

Keylogger: Case Study, [RedLine Stealer\(opens in new tab\)](#): RedLine Stealer acts as both a data stealer and a keylogger. It spreads through phishing or malicious websites, then captures keystrokes, browser cookies, and cryptocurrency wallet information.

C2 RAT Malware: Case Study, [\(opens in new tab\)](#)[QakBot\(opens in new tab\)](#): QakBot, also known as QBot, has become one of the most active and dangerous malware families targeting organisations globally. Originally designed as a banking trojan, QakBot has evolved into a modular RAT used by cybercriminal groups for initial access, credential theft, and as a loader for ransomware like Black Basta.

Task 4 - Binary vs Script Malware

Malware can arrive in different forms. Two of the most common categories are binary malware (compiled executables) and script-based malware (written in scripting languages).

Binary Malware: Executable-based malware is typically delivered as compiled files such as **.exe (Windows Executable)** or **.dll (Dynamic Link Library)**. These files are created from source code and need to be run on a compatible operating system. Common delivery methods for binaries include email attachments (for example, **invoice.pdf.exe**), malicious downloads, removable media, or as part of another infection.

Script-based Malware: Script-based malware relies on scripts written in languages such as **JavaScript (.js)**, **Visual Basic Script (.vbs)**, **Windows**

batch (.bat), or **PowerShell** (.ps1). Scripts can be easier for attackers to change and adapt. They often arrive as email attachments, within Office documents containing macros, or through malicious websites.

Task 5 - Practical

Suspicious process "ads_service.exe" spawned by user session. Multiple outbound HTTP requests to advertising domains detected.(ads_service.exe > adverd alert)

Unusual process "system_update.exe" accessing clipboard data and reading local email files. Network traffic shows regular uploads of small data packets to external IP 144.12.39.111. (regular not one time, small data > spyware alert)

Large volume of file modifications detected within shared directory "\\Server\Finance". File extensions changed from .docx to .encrypted. Ransom note text files discovered in multiple folders. (encrypted large volume file > ransomware alert)

Critical process "disk_cleaner.exe" executed with administrative privileges. Rapid deletion and overwriting of user files observed. System recovery points and backups also erased within short timeframe. (wiper malware)

Outbound connection initiated by "svchost32.exe" to previously unseen external host on port 443. Periodic beaconing detected every 30 minutes, no associated legitimate application found. (C2C)

User process "docu_manager.exe" accessed multiple sensitive files in "C:\Users\Public\Documents" and attempted to send large data upload to remote server 78.221.111.202. (access large data > data stealer)

Suspicious DLL injection detected in "winlogon.exe" process. Unauthorised API calls to keyboard input functions, along with repeated attempts to connect to remote IP 185.20.56.33. (login your stuff by keyboard > keylogger)

High CPU usage detected on host DESKTOP-07A2. Process "svcupdate.exe" utilising over 80% CPU for extended period. Multiple connections to known mining pool domains identified. (high CPU usage > Cryptominer)

Task 6 - Conclusion

Malware continues to be one of the most frequent threats defenders face, and understanding how to classify it helps improve detection and response.

Throughout this room, different types of malware were introduced, including how they behave and appear in real-world incidents and how security teams can identify and categorise them. Users learned to distinguish between malware families such as spyware, ransomware, wipers, and keyloggers, and reviewed real case studies to help build intuition. The differences between binary and script-based malware were also covered, along with common delivery methods and indicators of compromise.

Key takeaways:

- Learn how to identify and classify common types of malware.
- Understand how malware behaves and spreads in real-world incidents.
- Distinguish between binary and script-based malware formats.
- Apply classification skills using alert analysis in a simulated environment.